

# Activating Your Training: The Short-Range Plan

The short-range plan is where the rubber meets the road. It is what most people refer to as their daily schedule, although some will refer to it as a single “block” of training. This schedule lays out the specific workouts and rest periods for the days and weeks to come. Your short-range plan contains the precise volume, intensity, intervals, recovery, and terrain specificity on a day-by-day basis.

Many coaches and athletes arrange their short-range plans in tidy four-week periods of three weeks “on” and one week “off.” Generic twelve-week plans available in magazines or online are often arranged in this fashion, and while this offers a convenient way to organize a typical four-week month, I’ve always found this approach to be lazy and uneducated. The fact of the matter is that various stressors affect the body differently. It takes a different amount of time to recover from high volume compared to high intensity. Similarly, your body adapts along a different time course for various intensities. You cannot pigeonhole the process into rigid four-week time frames for every single training phase.

The 3:1 work-rest paradigm is a good reference point, but to get the most from your hard-earned training, this time frame needs to be adjusted to suit the adaptation you are seeking. The question is: How do you know what to adjust and in what ways? I use three primary concepts to design short-range plans, and when you apply these concepts to your training and your goals, you’ll find that your training better suits your individual needs, is more precise, and most likely breaks the 3:1 paradigm.

## Concepts for the Short-Range Plan

### **DO THE BIGGEST TRAINING LOAD WHEN YOU ARE THE MOST RESTED.**

You run the best when you are fresh. This is why we taper before races (though tapering tends to drive ultrarunners stir crazy) and will often have our best workouts right after a rest week. When we are fresh and rested, we can run faster for longer periods of time. If we extend this concept to training, you are the most ready to handle the biggest training load right after a recovery phase. Following that strategy, your short-range plan should start with the workout(s) with the biggest training load. Regardless of the phase you are in, you should aim to do the hardest workout you will do for the phase right off the bat. For example, for athletes who can handle a maximum of sixty minutes of SteadyStateRun, I will have them do that workout at the beginning of the phase, when they are the most rested. They might not be able to do a workout of that caliber for the remaining weeks of the phase, but that is fine. Many times, the hardest workout is actually a “B” race. You race hard, and the subsequent training stimulus of the race is far bigger than the stimulus you can achieve on any of the remaining training days. In any case, the fatigue induced from the initial few workouts will eventually lead to the desired adaptation. In this way, you must accept the fact that during the training process, you will get worse before you get better. If you do the right things in training, this is intentional (Figure 12.1).

| MON      | TUES                                                                                | WED                                                                                 | THURS                                                                               | FRI                                                                                 | SAT                                                                                 | SUN               |
|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------|
| 5        | 6                                                                                   | 7                                                                                   | 8                                                                                   | 9                                                                                   | 10                                                                                  | 11                |
| Rest Day | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 2:00 EnduranceRun |
| 12       | 13                                                                                  | 14                                                                                  | 15                                                                                  | 16                                                                                  | 17                                                                                  | 18                |
| Rest Day | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 4 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 EnduranceRun                                                                   | 2:00 EnduranceRun |
| 19       | 20                                                                                  | 21                                                                                  | 22                                                                                  | 23                                                                                  | 24                                                                                  | 25                |
| Rest Day | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 4 x 3 min RunningIntervals, 3 min recovery between intervals | 0:45 RecoveryRun                                                                    | 0:45 RecoveryRun                                                                    | 1:00 RecoveryRun                                                                    | 1:00 RecoveryRun  |

**FIGURE 12.1 A RunningIntervals phase where the hardest workouts are the first four.**

## WEEKLY PROGRESSIVE OVERLOADS HAVE CONSEQUENCES

**TAKE THE TRAINING PROGRAM IN FIGURE 12.1 and flip it around.** I see this type of training architecture time and time again, even with elite athletes. Each week builds off of the next with progressively more and more time at intensity. There is a huge glaring issue when athletes do this, and one that I think goes underappreciated in the endurance world. When athletes apply this type of weekly progressive workload training model, they end up attempting the hardest and/or longest workouts when they are most fatigued. This is a huge problem and a mistake many athletes and coaches make. It can be a costly mistake as well, since doing the hardest workout last can lead to insufficient recovery, maladaptation, or injury.

**HIGHER INTENSITY MEANS A SHORTER ADAPTATION PROCESS, LONGER RECOVERY, AND LOWER WORKOUT DENSITY.**

Most ultrarunners are intensity-averse, which means they do anything they can to avoid running very, very hard. If you are among the ultrarunning crowd that fears high-intensity work, the type of work that makes your lungs sear and your legs burn, the next paragraph will be a relief.

One of the saving graces of high-intensity work is that it takes very little of it to produce an adaptation. Hooray! The time you work at that intensity during any particular workout and the total number of consecutive weeks necessary to produce an adaptation are relatively small compared to the amount of moderate and lower-intensity work. Because of this, I never hesitate to work with “time-crunched” ultrarunners, because I can take their most time-crunched periods and still get excellent adaptations at high intensities. You can see that concept reflected in Figures 8.2 and 9.1, which show the time course for training adaptation.

But wait, there's more! It takes longer to recover from high-intensity work, which means that, workout for workout, you need more rest and recovery between sessions to get your body ready for the next hard session. From a practical standpoint, most high-intensity phases, particularly RunningIntervals, need to be only two or three weeks long to elicit an appropriate adaptation. So, the total number of "hard" workouts is markedly smaller because the length of the phase is shorter and the density of the hard workouts is lower. See Figure 12.2A (SteadyStateRun) and Figure 12.2B (RunningIntervals) for a comparison between a higher-intensity phase and a lower-intensity phase. SteadyStateRun is at a lower intensity, and therefore that phase can be longer than a RunningInterval phase.

## **LOWER INTENSITY MEANS A LONGER ADAPTATION PROCESS, SHORTER RECOVERY, AND HIGHER WORKOUT DENSITY.**

Workouts like EnduranceRuns, SteadyStateRuns, and to a lesser extent TempoRuns are typically at low enough intensities that it takes many weeks to accumulate enough training stress to produce an adaptation. Most ultra-athletes intuitively realize this, as their day-to-day training is mainly a compilation of EnduranceRuns and nausea. It is important to realize that even at the slightly higher intensities of SteadyStateRun and TempoRun, this can still be the case. Given the right construct, you should be able to handle back-to-back SteadyStateRun and TempoRun days for at least some of the phase and be able to stay in these phases for more than three weeks. At the TempoRun intensity, I have my athletes do these workouts two to three times per week, often scheduling two of these workouts on consecutive days. For SteadyStateRuns, that workout density gets pushed to four times per week in certain weeks.

During the long-range planning process, I always alert my athletes to what the hardest phase will be. The purpose is to better prepare them for what is to come. During that phase it is important to focus on recovery, get proper sleep, try to reschedule big work projects, and avoid other life stressors. If they are trying to change body composition, it is best to do it at some other point in the season. During this hardest phase, all of their spare energy will be needed for hard training and purposeful recovery. Ninety percent of the time this is the TempoRun, or lactate threshold, phase. Although this phase is not the most intense and does not have the biggest volume, it hits the right balance of volume and intensity. That balance tends to grind an athlete down over the course of time more so than higher intensities (with less volume) or lower intensities (with higher volume).

| A | MON      | TUES                                                                               | WED              | THURS                                                                              | FRI              | SAT                                                                                | SUN               |
|---|----------|------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------|-------------------|
|   | 5        | 6                                                                                  | 7                | 8                                                                                  | 9                | 10                                                                                 | 11                |
|   | Rest Day | 1:45 EnduranceRun with 2 x 30 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:45 EnduranceRun with 2 x 20 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 2:00 EnduranceRun with 2 x 30 min SteadyStateRun, 5 min recovery between intervals | 3:00 EnduranceRun |
|   | 12       | 13                                                                                 | 14               | 15                                                                                 | 16               | 17                                                                                 | 18                |
|   | Rest Day | 1:45 EnduranceRun with 2 x 30 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:45 EnduranceRun with 2 x 20 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 2:00 EnduranceRun with 2 x 20 min SteadyStateRun                                   | 3:30 EnduranceRun |
|   | 19       | 20                                                                                 | 21               | 22                                                                                 | 23               | 24                                                                                 | 25                |
|   | Rest Day | 1:45 EnduranceRun with 2 x 20 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:45 EnduranceRun with 1 x 30 min SteadyStateRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:30 EnduranceRun with 1 x 30 min SteadyStateRun                                   | 4:00 EnduranceRun |
|   | 26       | 27                                                                                 | 28               | 29                                                                                 | 30               | 31                                                                                 | 1                 |
|   | Rest Day | 1:00 RecoveryRun                                                                   | 1:00 RecoveryRun | 1:00 RecoveryRun                                                                   | 1:00 RecoveryRun | 2:00 EnduranceRun with 2 x 30 min SteadyStateRun, 5 min recovery between intervals | 3:00 EnduranceRun |

| B | MON      | TUES                                                                                | WED                                                                                 | THURS                                                                               | FRI                                                                                 | SAT                                                                                 | SUN               |
|---|----------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------|
|   | 5        | 6                                                                                   | 7                                                                                   | 8                                                                                   | 9                                                                                   | 10                                                                                  | 11                |
|   | Rest Day | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 2:00 EnduranceRun |
|   | 12       | 13                                                                                  | 14                                                                                  | 15                                                                                  | 16                                                                                  | 17                                                                                  | 18                |
|   | Rest Day | 1:30 EnduranceRun with 5 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 4 x 3 min RunningIntervals, 3 min recovery between intervals | 1:00 RecoveryRun                                                                    | 2:00 EnduranceRun |
|   | 19       | 20                                                                                  | 21                                                                                  | 22                                                                                  | 23                                                                                  | 24                                                                                  | 25                |
|   | Rest Day | 1:00 RecoveryRun                                                                    | 1:30 EnduranceRun with 4 x 3 min RunningIntervals, 3 min recovery between intervals | 0:45 RecoveryRun                                                                    | 0:45 RecoveryRun                                                                    | 1:00 RecoveryRun                                                                    | 1:00 RecoveryRun  |

**FIGURE 12.2 (A) A typical SteadyStateRun phase; (B) a typical RunningInterval phase. Note that the SteadyStateRun phase is longer and includes less recovery between the workouts than the RunningInterval phase.**

| MON      | TUES                                                                         | WED                                                                          | THURS                                                                       | FRI                                                                          | SAT                                                                          | SUN                                                                         |
|----------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 5        | 6                                                                            | 7                                                                            | 8                                                                           | 9                                                                            | 10                                                                           | 11                                                                          |
| Rest Day | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals  | 1:00 RecoveryRun                                                            | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals |
| 12       | 13                                                                           | 14                                                                           | 15                                                                          | 16                                                                           | 17                                                                           | 18                                                                          |
| Rest Day | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals  | 2:00 EnduranceRun                                                           |
| 19       | 20                                                                           | 21                                                                           | 22                                                                          | 23                                                                           | 24                                                                           | 25                                                                          |
| Rest Day | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals  | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals |
| 26       | 27                                                                           | 28                                                                           | 29                                                                          | 30                                                                           | 31                                                                           | 1                                                                           |
| Rest Day | 1:00 RecoveryRun                                                             | 1:00 RecoveryRun                                                             | 1:00 RecoveryRun                                                            | 1:30 EnduranceRun with 3 x 12 min TempoRun, 6 min recovery between intervals | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 2:00 EnduranceRun                                                           |

**FIGURE 12.3 A back-to-back-style training plan. There is one additional hard workout, as compared to Figure 12.4. Even in this example, the hardest workouts are still early in the phase.**

| MON      | TUES                                                                         | WED              | THURS                                                                       | FRI                                                                          | SAT                                                                          | SUN                                                                          |
|----------|------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 5        | 6                                                                            | 7                | 8                                                                           | 9                                                                            | 10                                                                           | 11                                                                           |
| Rest Day | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 2:00 EnduranceRun                                                            |
| 12       | 13                                                                           | 14               | 15                                                                          | 16                                                                           | 17                                                                           | 18                                                                           |
| Rest Day | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 1:00 RecoveryRun | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals | 2:00 EnduranceRun                                                            |
| 19       | 20                                                                           | 21               | 22                                                                          | 23                                                                           | 24                                                                           | 25                                                                           |
| Rest Day | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals  | 1:00 RecoveryRun | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 8 min TempoRun, 4 min recovery between intervals  | 2:00 EnduranceRun                                                            |
| 26       | 27                                                                           | 28               | 29                                                                          | 30                                                                           | 31                                                                           | 1                                                                            |
| Rest Day | 1:00 RecoveryRun                                                             | 1:00 RecoveryRun | 1:00 RecoveryRun                                                            | 1:30 EnduranceRun with 3 x 12 min TempoRun, 6 min recovery between intervals | 1:00 RecoveryRun                                                             | 1:30 EnduranceRun with 3 x 10 min TempoRun, 5 min recovery between intervals |

**FIGURE 12.4 A non-back-to-back-style training example. There are only nine hard workouts, as compared to the ten in the back-to-back style.**

## The Case for Back-to-Back Hard Days

Should you run on tired legs? Yes, but not for the reason you might think. Many ultrarunners have adopted a strategy with their long runs in which they intentionally run long and far on consecutive days. The theory is that this type of training teaches your body to “run when it is tired,” mimicking the demands of an ultramarathon. I adopt the back-to-back training strategy with my athletes in nearly every phase, so needless to say, I’m a fan. However, I take it a step further. Although I have my athletes do back-to-back long runs, I also have them do back-to-back days with SteadyStateRuns, TempoRuns, and, in certain circumstances, RunningIntervals. While using this strategy, you will certainly be more tired on the second day than the first. Your legs will feel like bricks, and if you have done the first workout correctly, you can’t perform quite up to the level that you did on Day 1.

But I do not have my athletes do back-to-back days to teach them to “run on tired legs.” I do it because the effect on the aerobic engine is greater when you are concentrate your training load. In addition, at times back-to-back days allow you to fit in one or two more high-quality workouts per phase (compare the examples in Figures 12.3 and 12.4). This is important because athletes tend to look at training from the standpoint of weekly volume, but you also need to consider your total workload for an entire phase of training. When you can schedule an extra workout or two within a period while still maintaining appropriate amounts of rest, you can wring greater gains out of that period of time. In practice, much of this strategy is adopted from other sports such as cross-country skiing and cycling (Rønnestad, Hansen, and Ellefsen 2014; Rønnestad et al. 2015). Even though running is a

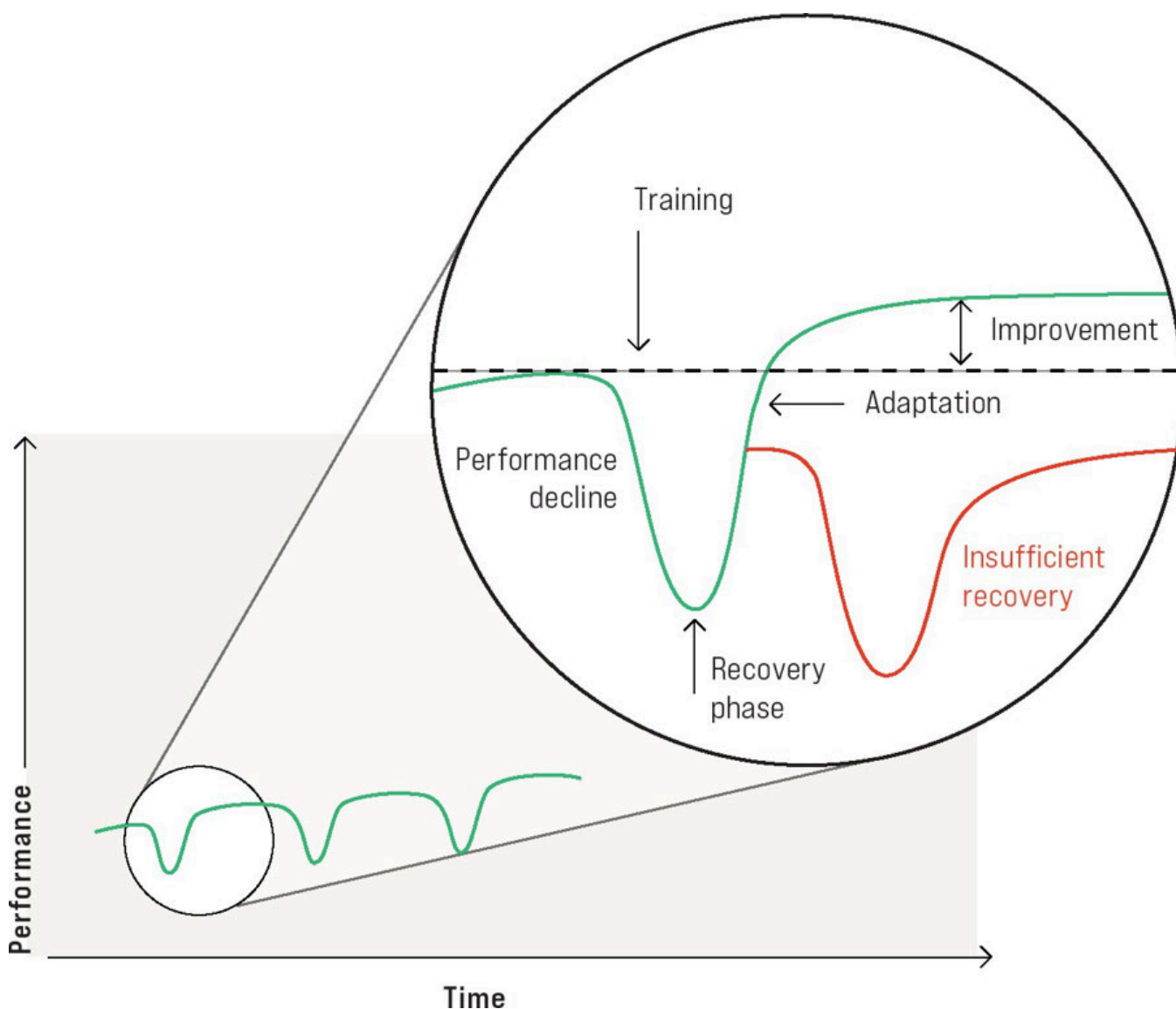


different sport and should be treated as such, many of the same principles apply across all aerobic sports. Taking this into account, I've adopted this block-style, load-concentration training strategy for the athletes I work with. Without a doubt, it's tricky to get right, and it requires careful monitoring of fatigue and performance. If you are designing your own training, I encourage you to first try back-to-back workouts at the SteadyStateRun intensity to see how you react. Do it once in a phase, preferably at the beginning. If you react well, you can try it across the course of the entire phase and then try it at different intensities.

## How to Schedule Recovery Phases

By now, you have almost all of the tools to design your own training plan. You have laid out your LRP in accordance with the principles in Chapter 10. Your short-range plan is nice and tidy and awaiting the next workout. Now you have to answer the questions “When is enough, enough?” and “When do I need to recover?”

You have likely heard time and time again that “you don't get better until you rest.” While this is true, the first part of the story is that you have to do the work—and you have to do enough of it to make a difference. That work, followed by an adequate recovery phase, should be enough to make small incremental improvements in your fitness. So, what you should really be asking yourself is: “When have I done enough work?”



## **FIGURE 12.5 How fitness (green line) is affected over time from the applications of training stress and recovery.**

To answer that question, I've developed a three-point checklist I work through with every athlete when determining when to apply rest. It's based on physiology, practice, and, if available, wearable devices that can track things like sleep and HRV.

### **TRAINING MEANS YOUR PERFORMANCE WILL DETERIORATE BEFORE YOU GET BETTER**

Before getting to the checklist, we all need to understand that, fundamentally, training makes your performance deteriorate in the short term (days and weeks) before you get better in the long term (weeks and months). You have to train hard enough that your body says, "Dang, this is hard. I better build up some more physiological stuff if I want to keep doing this and not die." Without a difficult-enough training stress and then the requisite recovery, you won't improve. You have to have both, not simply one or the other.

In this sense training can look like a good stock market chart with "performance" on the y-axis and "time" on the x-axis. The general trend over long periods of time will be upward, but it is certainly not trending up every single week.

The take-home message here is that you can't set PRs every week. If you are, congratulations! You are either new to the sport and/or your training is too easy and will stagnate at some point. Even if you have a moderate amount of training background, you should fully expect your performance to get moderately worse in some way before you apply rest and see further improvements.

### **POINT 1: PERFORMANCE DECLINES**

The first and most powerful indicator that you need to rest is when your daily performance starts to decline past a certain threshold. If you are regularly doing intervals and harder runs, you will have some basis of performance to benchmark week to week. By comparing the Normalized Graded Pace or Grade Adjusted Pace for the intervals and/ or your weekly hard runs, you can see if they are getting better, getting worse, or staying the same. When your performance drops off by 3 to 8 percent on more than two out of the last three workouts, it's time to give yourself a rest phase.

Any performance drop of less than 3 percent is within normal day-to-day variation. I would not even consider that a drop in performance, just noise in the measurement, so you can continue to train. Similarly, any one day where your performance drops by 3 to 8 percent should be expected, and I would not think twice about continuing to train.

However, stack two out of three sequential workouts that show a marked decline in performance, and that is a good indicator you have enough cumulative fatigue to warrant a rest phase. This is about finding the sweet spot between enough stress to make your body adapt and not so much that it will take an undue time to recover.

### **POINT 2: WHEN YOU FEEL WORSE FOR THREE DAYS IN A ROW**



Besides performance decline indicating when you need to rest, how you feel day to day can also be an indicator of overall fatigue. Just like one difficult training day should not keep you from your next workout, one day where you feel worse than you did the day before should not deter you either. However, if you feel progressively worse during your runs for three days in a row, it's time to take some recovery.

So, how bad do you have to feel to classify it as a “worse” day? Unlike hard workouts where you will have consistent and quantifiable benchmarks to compare against, feeling “worse” for three days in a row is in the eye of the beholder. If you are tracking your RPE (which you should), this could mean that RPE is a point or two higher for a normal EnduranceRun effort. Or you can scroll back through your training comments (which you should also be logging) and look for how you described each workout. If your comments indicate that you are “sluggish,” “flat,” or any similar vocabulary, these are written descriptors that you are feeling run-down. In any case, don't try to find hard and fast metrics. If you simply feel worse during your runs for a few days in a row, that's enough of an indicator that you should bring some needed recovery into the mix.

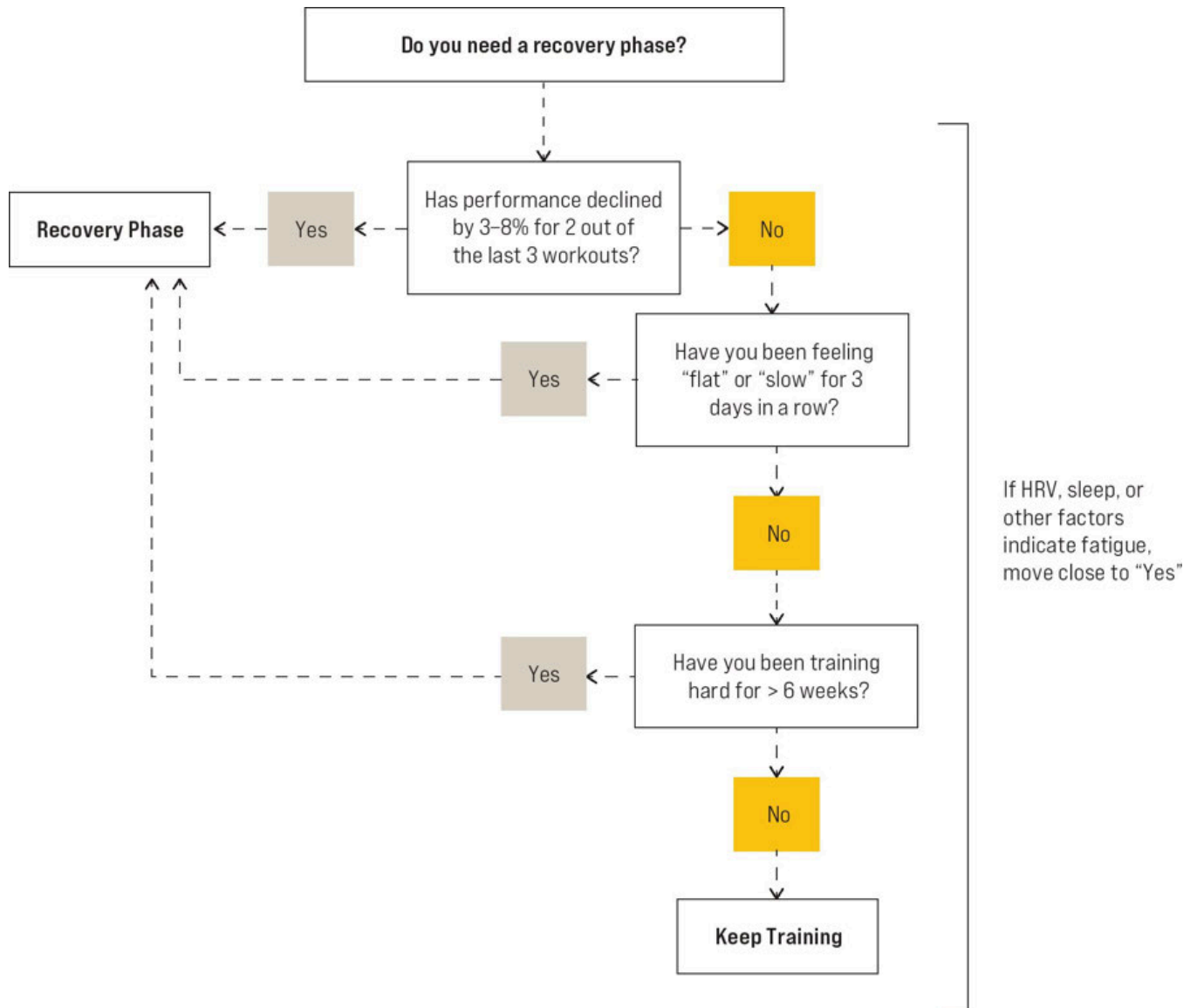
### **POINT 3: WHEN THE TRAINING PHASE GETS LONG**

Even in the absence of measurable performance declines or simply feeling bad, training phases can't last forever. But if your training is on fire, you feel good day in and day out, and you keep seeing improvements week to week, is there even a need to apply some recovery? The answer is yes.

It might be tempting to keep extending training phases in the absence of feeling fatigued or seeing a performance decline. But even in these circumstances, a rest phase reset should occur every so often. While there is no exact science to determine when to cap off a training block that seems to be going swimmingly, I only let training phases last for six weeks, maximum, before applying rest. The six-week maximum time frame is based a bit on experience as well as science. When researchers look at the time it takes for training to make an impact on performance, the general time frame is four to six weeks—meaning when you do a hard workout or series of hard workouts, it takes your body four to six weeks to harvest that work into some meaningful improvement in performance and/or physiology. So, incorporating rest near the end of this time frame allows you to take advantage of your body's natural buildup and breakdown processes.

### **WHAT ABOUT MONITORING HEART RATE VARIABILITY, RESTING HEART RATE, AND SLEEP CYCLES?**

More recently, various tools that measure physiological and lifestyle variables have become popular among athletes trying to gain a performance edge. Wearable technologies like the Whoop Strap and Oura Ring have emerged with a promise to determine when you are fatigued, whether your sleep is compromised, and when you should take a nap. I have seen and used these tools for decades, dating all the way back to taking a waking pulse on my wrist with no more technology than my fingers and a watch.



**FIGURE 12.6** Flowchart for determining whether it is time to incorporate a recovery phase.

While these new devices can add value to the training process, it's important to keep context in mind and not let the data tail wag the performance dog. Performance, or lack thereof, will still be your best indicator of when to apply rest. Information from wearables simply adds detail to the picture. This information should ever so slightly nudge you to "yes, I need to rest" or "no, I can keep training," but it should not drive the process.

## ARCHITECTURE OF THE RECOVERY PHASE

Don't overcomplicate this. Your recovery phase should consist of between four and six days of forty-five- to sixty-minute RecoveryRuns, with maybe an extra rest day thrown in for good measure. From the elite athletes I work with to the average Janes and Joes, this structure is almost universal. It's almost impossible to over-recover from runs being too short (duration) or workouts being too easy (intensity). You can, however, over recover if you take seven or more days easy, because beyond that time frame the benefit of more rest won't outweigh the gradual decay of your fitness. So, rest hard for several days, get your legs back underneath you, maybe catch up on some sleep, and get back after it once the recovery phase is over.

## **DON'T BE AFRAID TO REST EARLY**

When designing your short-range plan, realize that much of the time you will be guessing (in an educated way, of course). You will be forecasting how you will feel two to four or even six weeks from when you are creating your plan. Sometimes you are going to get that forecast right, and other times you will over- or undershoot how much training you can actually handle. Life adds nontraining stress to the mix in the form of working late, playing the soccer coach, and helping the kids out with schoolwork. Thus, you will also have to adapt to what is going on in real time.

When you are working through the aforementioned checklist, don't be afraid to rest early. Realize that your body can only handle so much stress over long periods of time, and it is usually advantageous to err on the side of resting early. If you slightly undershoot the length of the current training phase, you are going to give your body the ability to extend the next one. It will all come out in the wash as long as you are being reasonable phase to phase, working hard when you can, and backing off when you reach one of the three thresholds described above.

## **Tapering for Your Event**

Iñigo Mujika, a highly regarded sports physiologist known for his work on tapering and detraining, has defined tapering as “a progressive nonlinear reduction of the training load during a variable period of time, in an attempt to reduce the physiological and psychological stress of daily training and optimize sports performance” (Mujika and Padilla 2000a).

Contrary to what most popular literature with clickbait headlines will tell you, tapering will not make or break your race. On average, you stand to gain 3 percent from a properly constructed taper (Mujika and Padilla 2003). However, the difference is minimized when you consider that the 3 percent increase is from the baseline of implementing no taper at all. With simple, reasonable rest before your event, you might gain 2 percent, whereas if you designed the most scientific and effective taper ever, you would gain 3 percent. That's a difference of 1 percent, which is not much for most athletes. Yes, you should still implement a taper before your event. But you should also be realistic about what you expect to gain from it, be reasonable about the duration of your taper, and not worry too much if things don't go perfectly to plan. Remember, Mujika talks about reduction of physiological and psychological stress as goals of a taper, so stressing about achieving the perfect taper is counterproductive.

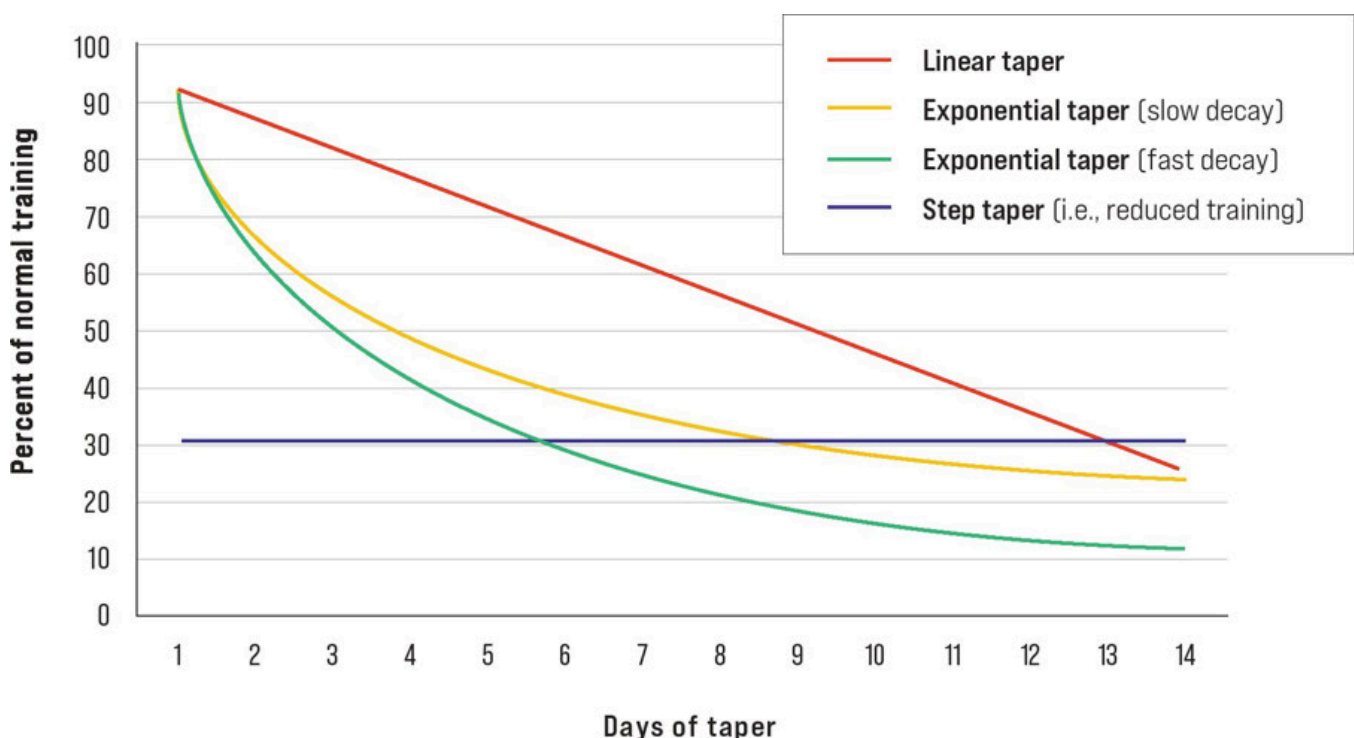
## **BEFORE YOU TAPER**

Train. One of the primary pitfalls of tapering occurs when athletes go into their goal event lacking confidence in their fitness. Remember Figure 10.4 and the time course for adaptation? Another point that chart illustrates is that after a period of time, if you are training properly, each subsequent workout results in fewer positive adaptations than the last. So, put in the work early. During the final two months of training, it's all about fine-tuning the fitness you have already built.

## **THE BASICS OF TAPERING**

The purpose of a taper is to simultaneously reduce the negative aspects of training (fatigue) and enhance the positive ones (fitness). This combination will leave you more fit and psychologically prepared for race day. The great thing is, by reducing the negative aspects of training, you naturally enhance the positive ones. From a strategic standpoint, this is crucial. Tapering fundamentally revolves around first reducing certain training variables; then, as a consequence of that reduction, adaptation takes place. In this way, tapering is not “maintenance,” which is a word that is carelessly thrown around (as in “I am going to maintain my fitness”). No athlete has ever come to me in my coaching career with “maintenance” as a goal. It should not be yours during the tapering process. Tapering is not maintenance. Tapering improves your physiology, leaving you more fit and ready to perform.

Training is a combination of volume, intensity, frequency, and environment. These four variables can be used to describe nearly every workout for an ultrarunner. When talking about tapering, it’s best to break it down into how these four variables should be manipulated. Before breaking tapering down by these variables, though, we need to talk about perspective. I love science. It provides a platform from which we have the opportunity to work and explore. The fundamentals provide the base, and we get to build on top of it. Every so often, though, science gets too caught up in the minutiae. Tapering is one of those areas.



**FIGURE 12.7 Schematic representation of the different types of tapers. Adapted from Mujika and Padilla 2003.**

If you review the best literature on tapering, the research will recommend a “nonlinear, fast-decay reduction” of training load. This reduction takes the shape of the green line shown in Figure 12.7, which illustrates two types of nonlinear reduction, fast and slow decay, as well as a step reduction (reducing training load by a set percentage all at once) and a linear reduction (reducing training load at a fixed percentage day after day). While I love the ingenuity of this graph and the countless hours spent on its development, this level of minutiae is impractical. For example, if you follow a “fast-decay” tapering process, on Day 5 your daily training volume is supposed to be at 32.5 percent. If you were normally doing a two-hour run, this means that your run for Day 5 should be thirty-nine minutes. Not forty or thirty-eight, but exactly thirty-nine. There’s nothing magical about thirty-nine minutes of running on Day 5 of a taper or any other day. Athletes are human beings, not machines or lab rats. In my experience, it’s best to understand the science and then find ways to use it practically.

Now, with a bit of perspective in hand, let’s look at the best overarching strategies for tapering based on the current literature. For the purposes of calibration, in this discussion of the components of training, I am assuming

a three-week tapering process for a runner who is training ten hours per week.

### **Volume: Reduce**

Adjusting volume is your first step in the tapering process. Using a fast-decay tapering model, volume should decrease very quickly at first and then level off toward the end. Thus, if you start with an overall volume of ten hours per week, you should reduce that to four to five hours (40 to 50 percent of normal training volume) per week the first week, two to three hours (20 to 30 percent of normal training volume) the second week, and two hours maximum (<20 percent of normal training volume) in the last week.

### **Intensity: Reduce the Amount, Not the Type**

Intensity is where things get a little tricky. The overarching strategy is to reduce the volume of intensity you do during the week but not the type of intensity. The amount that you reduce the intensity should follow the “fast-decay” strategy. Thus, similar to how you would reduce overall volume, you would reduce the volume of intensity quickly at first and then more slowly toward the end. For example, if in a normal training week you are doing four by ten-minute TempoRuns for a workout, during Taper Week 1 you would do three by five-minute TempoRuns (37.5 percent of the normal TempoRun volume). In Week 2 you would do two by five-minute TempoRuns (25 percent of the normal TempoRun volume), and in the last week one by five minutes (12.5 percent of the normal TempoRun volume).

Maintaining training intensity is a critical part of the tapering process. This is because the small amount of intensity that is done prevents a decline in the hard earned benefits you have worked for in training (Hickson et al. 1985; Sheply et al. 1992).

### **Frequency: Maintain or Slightly Reduce**

Frequency should be maintained or slightly reduced. This means that if you are running five days per week, continue running five days per week. If you are doing two specific workouts per week (TempoRun, SteadyStateRun, etc.), continue doing those two specific workouts per week using the guidelines above for reducing the volume of intensity for those workouts. If you decide to reduce the frequency of training, only do so by one day per week.

### **Environment: Maintain**

Whatever terrain and vertical specificity you have been doing in training leading up to a race, maintain that during the taper. This means if you have been running on trails, continue to run on trails. If you have been running on roads, run on roads. One of the mistakes I see over and over again is athletes who are targeting mountainous trail events train on specific trails for those events but then change that specificity for their taper. They run around the track in Chamonix in preparation for the Ultra-Trail du Mont-Blanc, for instance. This violates the final tapering strategy of specificity.

## THE TAPER TANTRUMS

The previously mentioned strategies are primarily (but not exclusively) aimed at improving your physiological state before an event. Your psychological state is just as important. And although the tapering strategies presented here have been demonstrated to also improve psychological qualities (Hooper, Mackinnon, and Howard 1999; Morgan et al. 1987; Raglin et al. 1996), many athletes still struggle with this aspect of race preparation. They feel as if they are losing fitness or missing out on training. The two to three weeks spent tapering seem to go on forever, and they doubt the process. They become antsy for the event, feel sluggish during their day-to-day life and runs, and become irritable and difficult to live with. This is normal, and it is important to understand that the “taper tantrums” are part of the process.

Angst and unease are normal psychological reactions to a taper. During the taper, you have more time and energy available than during your earlier training. That excess time and energy has to go somewhere, and many athletes utilize it simply to fret. To release their energy, they pack and repack their luggage, obsessively analyze splits, and second-guess their choice of socks. Any rational person observing these actions from the outside would probably get a chuckle, but these athletes aren't doing themselves any favors by expending their energy this way.

*To better handle your taper, trust your training. If you have done the right things in training, the taper becomes easier to handle. Have confidence that the miles, vertical, and intensity you have put in over the previous months are going to pay off. Even if your training has not gone perfectly, if you have been conscientious and dedicated to the process, you should trust that you have done enough.*

Create the right mindset for tapering. I routinely ask my athletes to review their training just in advance of their taper. Specifically I have them look at the sheer quantity of time they have dedicated to the process. The enormity of what they have done over the last several months puts the last few workouts in perspective. Those last workouts are a very, very small part of the entire process.

Avoid the taper tantrums by trusting your training and the tapering process. If you trust that you are going to come out the other side of a taper refreshed and ready for the event, the time you spend not running suddenly has purpose.

This is an excellent time to put your excess energy into rest. Yes, this takes just as much focus and energy as training, maybe even more so for ultrarunners. Take the time you had previously set aside for training and deliberately, intentionally set it aside for purposeful rest.

## KEY POINTS FOR SHORT-RANGE PLANNING

Do the hardest workouts when you are freshest. This is typically after a recovery week.

Consequentially, the volume of intensity for your workouts should decrease as the training phase progresses.

The higher the intensity of the training you are doing, the shorter any one phase can be.

RunningInterval phases can be two to four weeks followed by a four- to five-day recovery phase.

TempoRun phases can be three to four weeks followed by a four- to five-day recovery phase.

SteadyStateRun phases can be three to six weeks followed by a four- to five-day recovery phase.

Rest phases of four to six days should be incorporated when any of the following criteria are met:

- **Performance declines by 3 to 8 percent on two out of the last three workouts.**
- **You feel worse during runs on three consecutive days.**

You are in the same phase for:

- **four weeks if doing RunningIntervals**
- **four weeks if doing TempoRuns**
- **six weeks if doing SteadyStateRuns**

Start tapering two to three weeks out from an event.



- **Reduce your overall training load (overall volume and volume of intensity, but not the intensity itself) quickly at first, then gradually toward the end.**

- **Slightly reduce the frequency (days per week) or do not reduce the frequency at all.**

- **Maintain specificity (environment and running surface).**



